LAB 1

#1 WAP to display “Hello World” on console display

```
#include <iostream>
using namespace std;
int main(){
        cout<<"Hello World";
return 0;
}
```

```
hello world

...Program finished with exit code 0
Press ENTER to exit console.
```

#2 WAP that reads two values and displays larger value

```
#include <iostream>
using namespace std;
int main() {
        int
    num1,num2;
    cin>>num1>>num2;
    //using ternary operator
    (num1>num2) ? (cout<<"Larger is
"<<num1) : ((num2=num1) ?
(cout<<"Both are equal") :
(cout<<"Larger is "<<num2));
        return 0;
}
```

```
2 2
Both are equal

4 5
Larger is 5

5 4
Larger is 5
```

#3 WAP to read three values a,b,c, and display x = a/b-c

```
#include <iostream>
using namespace std;
int main() {
        float a,b,c;
    cin>>a>>b>>c;
    float x;
        x = a/b-c;
    cout<<x;
        return 0;
}
```

```
1 1 1
0

...Program finished with exit code 0
Press ENTER to exit console.
```

#4 WAP to implement control characters (\n, \t, \a, \r)

```
#include <iostream>
using namespace std;

int main() {
    cout<<"showing how to use \n to add new line and \t for tab space";
        cout<<"\n ring a bell using \a"<<endl;
    cout<<"carriage return brings cursor to \r beginning";

        return 0;
}
```

```
showing how to use \n
 to add new line and     \t for tab space
 ring a bell using \a
 beginningeturn brings cursor to \r

...Program finished with exit code 0
Press ENTER to exit console.
```

#5 Implement namespace in a program

```
#include <iostream>
using namespace std;
namespace one{
        int value = 100;
}
namespace two{
        int value=200;
}
int main(){
    cout<<"from first namespace: "<<one::value<<endl;
    cout<<"from second namespace: "<<two::value<<endl;
        return 0;
}
```

```
from first namespace: 100
from second namespace: 200


...Program finished with exit code 0
Press ENTER to exit console.
```

#6 WAP to define a structure student that contains 3 variables and 2 functions

```
#include <iostream>
using namespace std;
struct student{
        string
    name; int
    roll;
        int marks;

void setstudent(){
        cout << "Enter name: ";
        cin >> name;
        cout << "Enter roll number: ";
        cin >> roll;
        cout << "Enter marks: ";
        cin >> marks;
}
void getstudent(){
        cout << "\nName: " << name;
        cout << "\nRoll Number: " <<
    roll; cout << "\nMarks: " << marks <<
    endl;
}
};
int main(){
        student s[3];
        for (int i = 0; i < 3; i++){
                cout << "\nEnter details of
        student "
<< i + 1 << endl;
                s[i].setstudent();
        }
        for (int i = 0; i < 3; i++){
                cout << "\nDetails of student
        " << i
+ 1 << endl;
```

```
Enter details of student 1
Enter name: gauri
Enter roll number: 211
Enter marks: 94

Enter details of student 2
Enter name: aadhya
Enter roll number: 212
Enter marks: 95

Enter details of student 3
Enter name: mansha
Enter roll number: 213
Enter marks: 96

Details of student 1

Name: gauri
Roll Number: 211
Marks: 94

Details of student 2

Name: aadhya
Roll Number: 212
Marks: 95

Details of student 3

Name: mansha
Roll Number: 213
Marks: 96
```

#7 In the previous program make the data private and the functions public

```
#include <iostream>
using namespace std;
class student{
private:
        char
    name[50]; int
    roll; float
    marks;
public:
        void setstudent(){
                cout << "Enter name: ";
                cin >> name;
                cout << "Enter roll number:
        ";
                cin >> roll;
                cout << "Enter marks: ";
                cin >> marks;
        }
        void getstudent(){
                cout << "\nName: " << name;
                cout << "\nRoll Number: " <<
        roll; cout << "\nMarks: " << marks
        <<
endl;
    }
};
int main()
{
        student s[3];
        for (int i = 0; i < 3; i++){
        cout << "\nEnter details of student
" << i + 1 << endl;
                s[i].setstudent();
        }
        for (int i = 0; i < 3; i++){
                cout << "\nDetails of
```

```
Enter details of student 1
Enter name: gauri
Enter roll number: 1
Enter marks: 7

Enter details of student 2
Enter name: aadhya
Enter roll number: 2
Enter marks: 8

Enter details of student 3
Enter name: mansha
Enter roll number: 3
Enter marks: 9

Details of student 1

Name: gauri
Roll Number: 1
Marks: 7

Details of student 2

Name: aadhya
Roll Number: 2
Marks: 8

Details of student 3

Name: mansha
Roll Number: 3
Marks: 9
```

#8 Convert program #6 from struct to class

```
#include <iostream>
using namespace std;

class student {
private:
        string
    name; int
    roll; int
    marks;

public:
        void setstudent() {
                cout << "Enter name: ";
                cin >> name;
                cout << "Enter roll number:
        ";
                cin >> roll;
                cout << "Enter marks: ";
                cin >> marks;
        }

        void getstudent() {
                cout << "\nName: " << name;
                cout << "\nRoll Number: " <<
        roll; cout << "\nMarks: " << marks
        <<
endl;
    }
};

int main() {
        student s[3];

        for (int i = 0; i < 3; i++) {
        cout << "\nEnter details of student
" << i + 1 << endl;
```

```
Enter details of student 1
Enter name: g
Enter roll number: 1
Enter marks: 4

Enter details of student 2
Enter name: a
Enter roll number: 2
Enter marks: 5

Enter details of student 3
Enter name: m
Enter roll number: 3
Enter marks: 7

Details of student 1

Name: g
Roll Number: 1
Marks: 4

Details of student 2

Name: a
Roll Number: 2
Marks: 5

Details of student 3

Name: m
Roll Number: 3
Marks: 7
```

```
+ 1 << endl;
                s[i].getstudent();
        }

        return 0;
}
```

#9 Discuss the differences of structures in C and structures in C++

In C structures are simple data structures that are only used to hold and display data items, they don't contain functions. But in C++, structures contain functions.

Also in C++ the structure is treated as a class, we can have public/private members.

.

#10 WAP to read integer value and display on screen "WELL DONE" that many times.

```
#include <iostream>
using namespace std;

int main() {
        int x;
    cout<<"Enter how many times to
print: ";
        cin>>x;
            for(int i=0; i<x; i++)
        cout<<"WELL DONE"<<endl;
        return 0;
}
```

```
Enter how many times to print: 3
WELL DONE
WELL DONE
WELL DONE


...Program finished with exit code 0
Press ENTER to exit console.
```

LAB 2

#1 Create a structure in C++ containing the details of students as details below and a main function to execute the structure…

```
#include <iostream>
#include <string>
using namespace std;

    struct
    student{
    string name;
    int roll;
    string degree;
    char hostel;
    float cg;

        void adddetails()
        {
        cout<<"Enter
name,rollno,degree,hostel,cgpa: ";
                cin>>name>>roll>>degree>>hos
        tel>>cg;
        }
        void updatedetails()
        {
                adddetails();
        }
        void updatecgpa()
        {
                cout<<"Enter new cgpa: ";
                cin>>cg;
        }
            void updatehostel(){
        cout<<"Enter new hostel: ";
        cin>>hostel;
        }
            void displaydetails(){
        cout<<"\nName: "<<name;
        cout<<"\nRollNo: "<<roll;
```

```
PS C:\Users\lenovo\Desktop\OOPs> cd "c:\Users\
Enter no of students: 2
Enter details of student 1
Enter name,rollno,degree,hostel,cgpa: Gauri 1
Enter details of student 2
Enter name,rollno,degree,hostel,cgpa: Gauri 2

Student 1Details:

Name: Gauri
RollNo: 1
Degree: Civil
Hostel: A
Current CGPA: 9

Student 2Details:

Name: Gauri
RollNo: 2
Degree: Mech
Hostel: B
Current CGPA: 6.7
PS C:\Users\lenovo\Desktop\OOPs\a1>
```

```
                cout<<"\nCurrent CGPA:
        "<<cg<<endl;
        }
};

int main(){
        int n;
        cout<<"Enter no of students: ";
    cin>>n; student s[30];
        for(int i=0;i<n;i++){
        cout<<"Enter details of student
"<<i+1<<endl;
                s[i].adddetails();
        }
        for(int j=0;j<n;j++){
                cout<<"\nStudent
        "<<j+1<<"Details:
"<<endl;
                s[j].displaydetails();
        }
        return 0;
}
```

#2 Differentiate between public and private access/scope. Perform Q1 with class instead of structure with having the data members private and some member functions in private scope and some in public scope.

Public scope variables/functions can be accessed from anywhere in the program, by anyone. Whereas those that are private can be accessed within its specific class only.
Eg. Your bank details will be private but the functions of making payments, transfer etc are accessible to the public in a banking website.

```
#include <iostream>
#include <string>
using namespace std;

    class student{
    private:
        string
    name; int
    roll; string
    degree; char
    hostel; float
    cg;

        public:
        void adddetails()
        {
        cout<<"Enter
name,rollno,degree,hostel,cgpa: ";

cin>>name>>roll>>degree>>hostel>>cg;
        }
        void updatedetails()
        {
                adddetails();
        }
        void updatecgpa()
        {
                cout<<"Enter new cgpa: ";
                cin>>cg;
        }
            void updatehostel(){
        cout<<"Enter new hostel: ";
        cin>>hostel;
        }
            void displaydetails(){
```

```
Enter no of students: 2
Enter details of student 1
Enter name,rollno,degree,hostel,cgpa: ga
Enter details of student 2
Enter name,rollno,degree,hostel,cgpa: ma

Student 1Details:

Name: gauri
RollNo: 241
Degree: civil
Hostel: q
Current CGPA: 9.8

Student 2Details:

Name: mansha
RollNo: 234
Degree: enc
Hostel: q
Current CGPA: 9.6
```

```
        }
};

int main(){
        int n;
        cout<<"Enter no of students: ";
    cin>>n; student s[30];
        for(int i=0;i<n;i++){
        cout<<"Enter details of student
"<<i+1<<endl;
                s[i].adddetails();
        }
        for(int j=0;j<n;j++){
                cout<<"\nStudent
        "<<j+1<<"Details:
"<<endl;
                s[j].displaydetails();
        }
        return 0;
}
```

#3 Create a code snippet that illustrates the following: calling of private member functions inside public member function

```
#include <iostream>
using namespace std;
class xyz
{
        private:
        void SecretFn()
        {
        cout<<"welcome to the secret
function!";
        };
        public:
        void GetSecret()
        {
                SecretFn();
        }
};
```

```
welcome to the secret function!
PS C:\Users\lenovo\Desktop\DSA>
```

```
int main()
{
        xyz t;
    t.GetSecret();
}
```

#4 Define a class Rectangle with variables width and height of integer type along with following:

(a) void getdata() to initialize object values

(b) Void calculatearea() to calculate and display the area

```
#include <iostream>
using namespace std;

class Rectangle
{
        private:
        int width, height;

        public:
        void getData()
        {
                cout<<"Enter width and
        height: ";
                cin>>width>>height;
        }
        void calculateArea()
        {
        cout<<"Area =
"<<width*height<<endl;
        }
};
int main()
{
        Rectangle r1;
    r1.getData();
    r1.calculateArea();
}
```

```
Enter width and height: 2 4
Area = 8
PS C:\Users\lenovo\Desktop\DSA>
```

#5 Define a class Complex with variables real and imaginary along with following:
(a) void setComplex (float, float) to initialize object values.
(b) void displayComplex() to show the complex number
Pass and return objects to calculate sum of two complex numbers. Display the sum.

```
#include <iostream>
using namespace std;

class Complex {
private:
        float real;
    float imaginary;

public:
        void setComplex(float r, float i) {
                real =
        r; imaginary =
        i;
        }
            void
        displayComplex() { if
        (imaginary >= 0) {
            cout << real << " + " <<
imaginary << "i" << endl;
                } else {
                        cout << real << " -
            " <<
-imaginary << "i" << endl;
                }
        }

            Complex sum(Complex c1, Complex
        c2) { Complex temp;
                temp.real = c1.real +
        c2.real; temp.imaginary =
        c1.imaginary +
c2.imaginary;
                return temp;
        }
```

```
Enter real and imaginary parts of first comp
1 2
Enter real and imaginary parts of second co
3 4
First complex number: 1 + 2i
Second complex number: 3 + 4i
Sum of the two complex numbers: 4 + 6i
```

```
        float r1, i1, r2, i2;

    cout << "Enter real and imaginary parts
of first complex number: " << endl;
        cin
    >>r1;
    cin>>i1;
        c1.setComplex(r1, i1);
   cout << "Enter real and imaginary parts
of second complex number: " << endl;
        cin
    >>r2;
    cin>>i2;
        c2.setComplex(r2, i2);
        cout << "First complex number: ";
        c1.displayComplex();
        cout << "Second complex number: ";
        c2.displayComplex
    (); c3 = c3.sum(c1,
    c2);
    cout << "Sum of the two complex
numbers: ";
        c3.displayComplex
    (); return 0;
}
```

LAB 3

#1

```
#include <iostream>
using namespace std;

class Demo
{
    int x;
public:
    Demo(int x)
    {
        this->x = x;
    }
    void show()
    {
        cout<<this->x<<"\n";
    }
};

int main()
{
    Demo obj(10);
    Demo *p = &obj;

    obj.show();
    p->show();
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
10
10
```

#2

```
#include <iostream>
using namespace std;

class B;

class A
{
    int x;
```

```
PS C:\Users\lenovo\Desktop\DSA>
20 10
```

```
public:
        A(int x){ this->x=x; }
    friend void swap(A&,B&);
    void show(){ cout<<x<<" "; }
};

class B
{
    int y;
public:
        B(int y){ this->y=y; }
    friend void swap(A&,B&);
    void show(){ cout<<y<<" "; }
};

void swap(A &a,B &b)
{
        int
    t=a.x;
    a.x=b.y;
    b.y=t;
}

int main()
{
        A a(10);
        B b(20);
        swap(a,b);
        a.show();
        b.show();
}
```

#3

```
#include <iostream>
using namespace std;

class B;

class A
```

```
PS C:\Users
30
```

```
{
    int x;
public:
        A(int x){ this->x=x; }
    friend int add(A,B);
};

class B
{
    int y;
public:
        B(int y){ this->y=y; }
    friend int add(A,B);
};

int add(A a,B b)
{
        return a.x+b.y;
}

int main()
{
        A a(10);
        B b(20);
        cout<<add(a,b);
}
```

#4

```
#include <iostream>
using namespace std;

class A
{
    int x;
    friend class B;
public:
    A(int x){ this->x=x; }
};
```

```
PS C:\Users\lenovo\Desktop\DSA>
50
```

```
class B
{
public:
        void show(A a)
        {
                cout<<a.x;
        }
};

int main()
{
        A a(50);
        B b;
    b.show(a);
}
```

#5

```
#include <iostream>
using namespace std;

class Rect
{
    int l,b;
public:
    void set(int l,int b)
    {
        this->l=l
        ;
        this->b=b
        ;
    }
    int area()
    {
        return l*b;
    }
};

int main()
{
```

```
PS C:\Users\lenovo\Desktop\DSA>
 ; if ($?) { .\tempCodeRunnerFil
6 20 42
```

```
        Rect r[3];
    r[0].set(2,3);
        r[1].set(4,5);
        r[2].set(6,7);

            for(int i=0;i<3;i++)
        cout<<r[i].area()<<" ";
}
```

#6

```
#include <iostream>
using namespace std;

inline int cube(int x)
{
    return x*x*x;
}

int main()
{
    int n;
    cin>>n;
    cout<<cube(n);
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
 ; if ($?) { .\tempCodeRunnerFil
4
64
```

#7

```
#include <iostream>
using namespace std;

class A
{
public:
    int x;
    A(int x=0){ this->x=x; }
};

A modify(A a)
{
```

```
PS C:\Users\len
15
```

```
        a.x+=10;
        return a;
}

int main()
{
        A a(5);
        A
    b=modify(a);
    cout<<b.x;
}
```

LAB 4

#1

```
#include <iostream>
using namespace std;
class Rectangle
{
private:
    float length;
    float breadth;
public:
    Rectangle()
    {
        length = 0;
        breadth = 0;
    }
    Rectangle(float l, float b)
    {
        length = l;
        breadth = b;
    }
    Rectangle(float x)
    {
        length = x;
        breadth = x;
    }
    float area()
    {
        return length * breadth;
    }
};
int main()
{
    Rectangle r1;
    Rectangle r2(5);
    Rectangle r3(4, 6);
    cout << "Area of r1: " << r1.area() << endl;
    cout << "Area of r2: " << r2.area() << endl;
    cout << "Area of r3: " << r3.area() << endl;
    return 0;
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
Area of r1: 0
Area of r2: 25
Area of r3: 24
```

#2

```
#include <iostream>
using namespace std;

class Rectangle
{
private:
    float length;
    float breadth;

public:
    Rectangle(float l = 0, float b = 0)
    {
        length = l;
        breadth = b;
        cout << "Constructor called\n";
    }

    float area()
    {
        return length * breadth;
    }
    ~Rectangle()
    {
        cout << "Destructor called\n";
    }
};

int main()
{
    Rectangle arr[3] = { Rectangle(),
Rectangle(5), Rectangle(4,6) };

    for(int i = 0; i < 3; i++)
    {
        cout << "Area of object " << i+1 << ": "
<< arr[i].area() << endl;
    }

    return 0;
}
```

```
Constructor called
Constructor called
Constructor called
Area of object 1: 0
Area of object 2: 0
Area of object 3: 24
Destructor called
Destructor called
Destructor called
```

#3

```
#include <iostream>
using namespace std;
class Demo
{
public:
    Demo()
    {
        cout << "Constructor called\n";
    }
```

```
Constructor called
Destructor called
```

```
    ~Demo()
    {
        cout << "Destructor called\n";
    }
};
int main()
{
    Demo obj;
    return 0;
}
```

#4

```
#include <iostream>
using namespace std;

class Sample
{
public:
    Sample()
    {
        cout << "Object Created\n";
    }

    ~Sample()
    {
        cout << "Object Destroyed\n";
    }
};

int main()
{
    int *p = new int;
    *p = 10;
    cout << "Integer value: " << *p << endl;
    delete p;
    float *f = new float;
    *f = 5.5;
    cout << "Float value: " << *f << endl;
    delete f;
    int *arr = new int[3];
    arr[0] = 1;
    arr[1] = 2;
    arr[2] = 3;
    cout << "Integer array: ";
    for(int i = 0; i < 3; i++)
        cout << arr[i] << " ";
    cout << endl;
    delete[] arr;
    // Float array
    float *arr2 = new float[2];
    arr2[0] = 2.5;
    arr2[1] = 4.5;
    cout << "Float array: ";
    for(int i = 0; i < 2; i++)
```

```
PS C:\Users\lenovo\De
Integer value: 10
Float value: 5.5
Integer array: 1 2 3
Float array: 2.5 4.5
Object Created
Object Destroyed
Object Created
Object Created
Object Destroyed
Object Destroyed
```

```
        cout << arr2[i] << " ";
    cout << endl;
    delete[] arr2;
    // Single object
    Sample *obj = new Sample();
    delete obj;
    // Array of objects
    Sample *objArr = new Sample[2];
    delete[] objArr;
    return 0;
}
```

LAB 5

#1

```
#include <iostream>
using namespace std;

class Base
{
public:
    void showBase()
    {
        cout << "Base class function\n";
    }
};

class Derived : public Base
{
public:
    void showDerived()
    {
        cout << "Derived class function\n";
    }
};

int main()
{
    Base b;
    Derived d;

    b.showBase();
    d.showBase();
    d.showDerived();

    return 0;
}
```

```
Base class function
Base class function
Derived class function
```

#2

```
#include <iostream>
using namespace std;

class Base
{
protected:
    int x;
public:
    Base()
    {
        x = 10;
    }
};
```

```
Value of x: 10
```

```
class Derived : public Base
{
public:
    void display()
    {
        cout << "Value of x: " << x << endl;
    }
};

int main()
{
    Derived d;
    d.display();
    return 0;
}
```

#3

//PUBLIC:

```
#include <iostream>
using namespace std;

class A
{
public:
    int x = 5;
};

class B : public A
{
};

int main()
{
    B obj;
    cout << obj.x << endl;
    return 0;
}
```

```
PS C:\Use
5
PS C:\Use
5
PS C:\Use
5
```

//PROTECTED:

```
#include <iostream>
using namespace std;

class A
{
public:
    int x = 5;
};

class B : protected A
{
public:
    void show()
```

```
    {
        cout << x << endl;
    }
};

int main()
{
    B obj;
    obj.show();
    return 0;
}
```

//PRIVATE:

```
#include <iostream>
using namespace std;

class A
{
public:
    int x = 5;
};

class B : private A
{
public:
    void show()
    {
        cout << x << endl;
    }
};

int main()
{
    B obj;
    obj.show();
    return 0;
}
```

#4

//SINGLE

```
#include <iostream>
using namespace std;
class A
{
public:
    void showA()
    {
        cout << "Class A\n";
    }
};
class B : public A
{
};
```

```
PS C:\Users\lenovo\De
Class A
```

```
int main()
{
    B obj;
    obj.showA();
    return 0;
}
```

```
PS C:\Users\lenovo\De
Class A
Class B
```

//MULTIPLE:

```
#include <iostream>
using namespace std;
class A
{
public:
    void showA()
    {
        cout << "Class A\n";
    }
};
class B
{
public:
    void showB()
    {
        cout << "Class B\n";
    }
};
class C : public A, public B
{
};

int main()
{
    C obj;
    obj.showA();
    obj.showB();
    return 0;
}
```

```
Class A
```

//MULTILEVEL:

```
#include <iostream>
using namespace std;
class A
{
public:
    void showA()
    {
        cout << "Class A\n";
    }
};
class B : public A
{
};
class C : public B
{
```

```
};

int main()
{
    C obj;
    obj.showA();
    return 0;
}
```

//Hierarchical:

```
#include <iostream>
using namespace std;
class A
{
public:
    void show()
    {
        cout << "Base Class\n";
    }
};
class B : public A
{
};
class C : public A
{
};
int main()
{
    B b;
    C c;
    b.show();
    c.show();

    return 0;
}
```

```
PS C:\Users\le
Base Class
Base Class
```

//HYBRID:

```
#include <iostream>
using namespace std;
class A
{
public:
    void showA()
    {
        cout << "Class A\n";
    }
};
class B : public A
{
};
class C : public A
{
};
class D : public B, public C
```

```
Class A
Class A
```

```
{
};
int main()
{
    D obj;
    obj.B::showA();
    obj.C::showA();
    return 0;
}
```

#5

```
#include <iostream>
using namespace std;
class Base
{
public:
    Base()
    {
        cout << "Base Constructor\n";
    }
    ~Base()
    {
        cout << "Base Destructor\n";
    }
};
class Derived : public Base
{
public:
    Derived()
    {
        cout << "Derived Constructor\n";
    }
    ~Derived()
    {
        cout << "Derived Destructor\n";
    }
};

int main()
{
    Derived obj;
    return 0;
}
```

```
Base Constructor
Derived Constructor
Derived Destructor
Base Destructor
```

#6

```
#include <iostream>
using namespace std;
class Book
{
protected:
    string title;
    string author;
    float price;
public:
    Book(string t, string a, float p)
    {
        title = t;
        author = a;
        price = p;
    }
    void displayBook()
    {
        cout << "Title: " << title << endl;
        cout << "Author: " << author << endl;
        cout << "Price: " << price << endl;
    }
};

class Textbook : public Book
{
    string subject;

public:
    Textbook(string t, string a, float p, string s) :
Book(t, a, p)
    {
        subject = s;
    }

    void displayTextbook()
    {
        displayBook();
        cout << "Subject: " << subject << endl;
    }
};

int main()
{
    Textbook t("C++ Programming", "Bjarne",
500, "Programming");
    t.displayTextbook();
    return 0;
}
```

```
Title: C++ Programming
Author: Bjarne
Price: 500
Subject: Programming
```

#7

```
#include <iostream>
using namespace std;
class Speedometer
{
	protected:
		int speed;
	public:
		Speedometer(int s)
		{
			speed = s;
		}
		void showspeed()
		{
			cout<<"Speed =
"<<speed<<endl;
		}
};
class fuelgauge
{
	protected:
		int fuel;
	public:
		fuelgauge(int f)
		{
			fuel = f;
		}
		void showfuel()
		{
			cout<<"Fuel =
"<<fuel<<endl;
		}
};
class thermo
{
	protected:
		int temp;
	public:
		thermo(int t)
		{
			temp=t;
		}
		void showtemp()
		{
			cout<<"Temp =
"<<temp<<endl;
		}
};
class dash:  public Speedometer,public
fuelgauge, public thermo
{
	public:
		dash(int s,int f, int t) :
Speedometer(s),fuelgauge(f),thermo(t)
		{
```

```
Speed = 80
Fuel = 40
Temp = 30
```

```
		}
		void dispdash()
		{
			showspeed();
			showfuel();
			showtemp();
		}
};
int main()
{
	dash car(80,40,30);
	car.dispdash();
	return 0;
}
```

#8

```
#include <iostream>
using namespace std;
class libuser
{
	protected:
		string name;
		int id;
		string contact;
	public:
		libuser(string n, int i, string c)
		{
			name=n;
			id=i;
			contact=c;
		}
		void disp()
		{
			cout<<"Name:
"<<name<<endl;
			cout<<"ID: "<<id<<endl;
			cout<<"Contact:
"<<contact<<endl;
		}
};
class student:public libuser
{
	string grade;
public:
	student(string n, int i, string c, string g):
libuser(n,i,c)
	{
		grade = g;
	}
	void dispstud()
	{
		disp();
		cout<<"Grade: "<<grade<<endl;
	}
```

```
Name: Gauri
ID: 211
Contact: 9032023903
Grade: A

Name: abc
ID: 232
Contact: 93203092
Dept: maths
```

```
};
class teacher: public libuser
{
	string dept;
public:
	teacher(string n, int i, string c, string d) :
libuser(n,i,c)
	{
		dept=d;
	}
	void dispteach()
	{
		disp();
		cout<<"Dept: "<<dept<<endl;
	}
};
int main()
{
	student
s("Gauri",211,"9032023903","A");
	teacher
t("abc",232,"93203092","maths");
	s.dispstud();
	cout<<endl;
	t.dispteach();
	return 0;
}
```

#9

```
#include <iostream>
using namespace std;
class vehicle
{
	protected:
		string make;
		int year;
		string model;
	public:
		vehicle(string m, int y, string mo)
		{
			make=m;
			year=y;
			model=mo;
		}
		void disp()
		{
			cout<<"Make:
"<<make<<endl;
			cout<<"Model:
"<<model<<endl;
			cout<<"Year:
"<<year<<endl;
		}
};
```

```
Make: Tata
Model: xyz
Year: 2000
Load capactiy = 50000000
Temp control: -18

--------------------------------
```

```
class truck:public vehicle
{
protected:
	int loadcap;
public:
	truck(string m,int y,string mo,int lc):
vehicle(m,y,mo)
	{
		loadcap=lc;
	}
	void dispt()
	{
		disp();
		cout<<"Load capactiy =
"<<loadcap<<endl;
	}
};
class rt: public truck
{
	int tempc;
public:
	rt(string m,int y,string mo,int lc,int tc) :
truck(m,y,mo,lc)
	{
		tempc=tc;
	}
	void disprt()
	{
		dispt();
		cout<<"Temp control:
"<<tempc<<endl;
	}
};
int main()
{
	rt t1("Tata",2000,"xyz",50000000,-18);
	t1.disprt();
	return 0;
}
```

#10

```
#include <iostream>
using namespace std;
class person
{
	protected:
		string name;
		string address;
	public:
		person(string n,string a)
		{
			name=n;
			address=a;
		}
		void disp()
		{
			cout<<"Name:
"<<name<<endl;
			cout<<"Address:
"<<address<<endl;
		}
};
class staff:virtual public person
{
protected:
	int empid;
	string dep;
public:
	staff(string n,string a,int id,string d):
person(n,a)
	{
		empid=id;
		dep=d;
	}
	void dispstaf()
	{
		disp();
		cout<<"Employee id:
"<<empid<<endl;
		cout<<"Department:
"<<dep<<endl;
	}
};
class student: virtual public person
{
protected:
	int studid;
	string grade;
public:
	student(string n,string a,int id,string
g):person(n,a)
	{
		studid=id;
		grade=g;
	}
	void dispstud()
```

```
Name: ABC
Address: city
Name: ABC
Address: city
Employee id: 3902
Department: CSED
student id: 3920
grade: A
```

```
	{
		cout<<"student id:
"<<studid<<endl;
		cout<<"grade: "<<grade<<endl;
	}
};
class ta: public staff,public student
{
public:
	ta(string n,string a,int empid,string
dep,int sid,string
g):person(n,a),staff(n,a,empid,dep),student(n,a,
sid,g)
	{
	}
	void dispta()
	{
		disp();
		dispstaf();
		dispstud();
	}
};
int main()
{
	ta
ta1("ABC","city",3902,"CSED",3920,"A");
	ta1.dispta();
	return 0;
}
```

LAB 6

#1

```
#include<iostream>
#include<string>
using namespace std;

class Book
{
public:
    string title,author;
    long isbn;
};
class Library
{
    Book b[50];
    int count;

public:
    Library()
    {
        count=0;
    }
    bool addNewBook(string &title,string &author,long &isbn);
    bool removeBooks(long &isbn);
    void displayDetails();
};

bool Library::addNewBook(string &title,string &author,long
&isbn)
{
    b[count].title=title;
    b[count].author=author;
    b[count].isbn=isbn;
    count++;
    return true;
}

bool Library::removeBooks(long &isbn)
{
    for(int i=0;i<count;i++)
    {
        if(b[i].isbn==isbn)
        {
            for(int j=i;j<count-1;j++)
                b[j]=b[j+1];

            count--;
            cout<<"Removed Book";
            return true;
        }

    }
    return false;
}

void Library::displayDetails()
{
    cout<<"Details of books: "<<endl;
    for(int i=0;i<count;i++)
```

```
Enter title, author, isbn of 5 books
b1 a1 1
b2 a2 2
b3 a3 3
b4 a4 4
b5 a5 5
Now enter isbn of book to remove:
4
Removed BookDetails of books:
b1 a1 1
b2 a2 2
b3 a3 3
b5 a5 5
PS C:\Users\lenovo\Desktop\DSA>
```

LAB 7

#1

```
#include<iostream>
using namespace std;

class polygon
{
protected:
    int width,height;

public:
    void set_value(int w,int h)
    {
        width=w;
        height=h;
    }
};

class rectangle: public polygon
{
public:
    int calculate_area()
    {
        return width*height;
    }
};

class triangle: public polygon
{
public:
    int calculate_area()
    {
        return width*height/2;
    }
};

int main()
{
    polygon *p;
    rectangle r;
    triangle t;

    p=&r;
    p->set_value(10,5);
    cout<<"Rectangle area: "<<r.calculate_area()<<endl;

    p=&t;
    p->set_value(10,5);
    cout<<"Triangle area: "<<t.calculate_area()<<endl;
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
Rectangle area: 50
Triangle area: 25
```

#2

```
#include<iostream>
using namespace std;

class shape
{
public:
    virtual void area(){}
    virtual void display(){}
};

class circle: public shape
{
    float r;
public:
    circle(float x){r=x;}

    void area()
    {
        cout<<"Area of circle: "<<3.14*r*r<<endl;
    }

    void display()
    {
        cout<<"Shape: Circle"<<endl;
    }
};

class rectangle: public shape
{
    float l,b;
public:
    rectangle(float x,float y){l=x;b=y;}

    void area()
    {
        cout<<"Area of rectangle: "<<l*b<<endl;
    }

    void display()
    {
        cout<<"Shape: Rectangle"<<endl;
    }
};

class triangle: public shape
{
    float b,h;
public:
    triangle(float x,float y){b=x;h=y;}

    void area()
    {
        cout<<"Area of triangle: "<<0.5*b*h<<endl;
    }

    void display()
```

```
    {
        cout<<"Shape: Triangle"<<endl;
    }
};

int main()
{
    shape *s;

    circle c(5);
    rectangle r(4,6);
    triangle t(3,7);

    s=&c; s->display(); s->area();
    s=&r; s->display(); s->area();
    s=&t; s->display(); s->area();
}
```

```
Shape: Circle
Area of circle: 78.5
Shape: Rectangle
Area of rectangle: 24
Shape: Triangle
Area of triangle: 10.5
PS C:\Users\lenovo\Desktop\DSA>
```

#3

```
#include<iostream>
#include<cmath>
using namespace std;

float area(float b,float h)
{
    return 0.5*b*h;
}

float area(float a)
{
    return (1.732/4)*a*a;
}

float area(float a,float b,float c)
{
    float s=(a+b+c)/2;
    return sqrt(s*(s-a)*(s-b)*(s-c));
}

int main()
{
    cout<<"Right triangle: "<<area(4,5)<<endl;
    cout<<"Equilateral: "<<area(6)<<endl;
    cout<<"Isosceles: "<<area(5,5,6)<<endl;
}
```

```
Right triangle: 10
Equilateral: 15.588
Isosceles: 12
```

#4

```
#include<iostream>
using namespace std;

class Student
{
public:
    virtual void show()=0;
};

class Engineering: public Student
{
public:
    void show()
    {
        cout<<"Engineering student"<<endl;
    }
};

class Medicine: public Student
{
public:
    void show()
    {
        cout<<"Medicine student"<<endl;
    }
};

class Science: public Student
{
public:
    void show()
    {
        cout<<"Science student"<<endl;
    }
};

int main()
{
    Student *s[3];

    Engineering e;
    Medicine m;
    Science sc;

    s[0]=&e;
    s[1]=&m;
    s[2]=&sc;

    for(int i=0;i<3;i++)
        s[i]->show();
```

```
}
```

```
Engineering student
Medicine student
Science student
```

#5

```
#include<iostream>
using namespace std;

class Time
{
    int h,m,s;

public:
    Time(){}
    Time(int a,int b,int c)
    {
        h=a; m=b; s=c;
    }

    Time operator+(Time t)
    {
        Time temp;
        temp.s=s+t.s;
        temp.m=m+t.m+temp.s/60;
        temp.s%=60;

        temp.h=h+t.h+temp.m/60;
        temp.m%=60;

        return temp;
    }

    void show()
    {
        cout<<h<<":"<<m<<":"<<s<<endl;
    }
};

int main()
{
    Time t1(5,15,34),t2(9,53,58),t3;
    t3=t1+t2;
    t3.show();
}
```

```
15:9:32
PS C:\Users\lenovo\Desktop\DSA>
```

#6

```
#include<iostream>
#include<string.h>
using namespace std;

class STRING
```

```
{
    char s[100];

public:
    void input()
    {
        cin>>s;
    }

    bool operator==(STRING t)
    {
        return strcmp(s,t.s)==0;
    }

    STRING operator+(STRING t)
    {
        STRING temp;
        strcpy(temp.s,s);
        strcat(temp.s,t.s);
        return temp;
    }

    void display()
    {
        cout<<s<<endl;
    }
};

int main()
{
    STRING s1,s2,s3;

    s1.input();
    s2.input();

    if(s1==s2)
        cout<<"Equal"<<endl;
    else
        cout<<"Not Equal"<<endl;

    s3=s1+s2;
    s3.display();
}
```

```
PS C:\Users\lenovo\Desktop\DSA
helloo
hellooo
Not Equal
helloohellooo
```

#7

```
#include<iostream>
using namespace std;

class Matrix
{
```

```
    int a[10][10],r,c;

public:
    void input()
    {
        cout<<"Enter rows and cols: ";
        cin>>r>>c;

        for(int i=0;i<r;i++)
            for(int j=0;j<c;j++)
                cin>>a[i][j];
    }

    void display()
    {
        for(int i=0;i<r;i++)
        {
            for(int j=0;j<c;j++)
                cout<<a[i][j]<<" ";
            cout<<endl;
        }
    }

    friend Matrix operator*(Matrix,Matrix);
};

Matrix operator*(Matrix m1,Matrix m2)
{
    Matrix m3;
    m3.r=m1.r;
    m3.c=m2.c;

    for(int i=0;i<m3.r;i++)
    {
        for(int j=0;j<m3.c;j++)
        {
            m3.a[i][j]=0;
            for(int k=0;k<m1.c;k++)
                m3.a[i][j]+=m1.a[i][k]*m2.a[k][j];
        }
    }
    return m3;
}

int main()
{
    Matrix m1,m2,m3;

    cout<<"Matrix 1"<<endl;
    m1.input();

    cout<<"Matrix 2"<<endl;
    m2.input();

    m3=m1*m2;

    cout<<"Result"<<endl;
```

```
    m3.display();
}
```

```
Matrix 1
Enter rows and cols: 2 2
1 0 1 0
Matrix 2
Enter rows and cols: 2 2
1 1 1 1
Result
1 1
1 1
```

#8

```
#include<iostream>
using namespace std;

class Array
{
    int a[10];

public:
    void input()
    {
        for(int i=0;i<10;i++)
            cin>>a[i];
    }

    int operator[](int index)
    {
        if(index<0 || index>=10)
        {
            cout<<"Index out of bounds"<<endl;
            return -1;
        }
        return a[index];
    }
};

int main()
{
    Array arr;
    arr.input();
    int i;
    cout<<"Enter index: ";
    cin>>i;
    cout<<arr[i];
}
```

```
1 2 3 4 5 6 8 9 0 7
Enter index: 4
5
PS C:\Users\lenovo\Desktop\
1 2 3 4 5 6 7 8 9 0
Enter index: 11
Index out of bounds
-1
```

#9

```
#include<iostream>
using namespace std;

class Test
{
public:
    void operator()(int a,int b,int c)
    {
        cout<<"Sum = "<<a+b+c<<endl;
    }
};

int main()
{
    Test t;
    t(10,20,30);
}
```

```
PS C:\Users\lenovo\Desktop\DSA
Sum = 60
```

#10

```
#include<iostream>
using namespace std;

class Test
{
    int x;
public:
    friend istream& operator>>(istream &in,Test &t);
    friend ostream& operator<<(ostream &out,Test &t);
};

istream& operator>>(istream &in,Test &t)
{
    in>>t.x;
    return in;
}

ostream& operator<<(ostream &out,Test &t)
{
    out<<t.x;
    return out;
}
```

```
int main()
{
    Test t;
    cout<<"Enter value: ";
    cin>>t;
    cout<<"Value: "<<t;
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
Enter value: 56
Value: 56
```

#11

```
#include<iostream>
using namespace std;

class Test
{
    float x;

public:
    Test(float f)
    {
        x=f;
    }

    void show()
    {
        cout<<x;
    }
};

int main()
{
    float f;
    cout<<"Enter float: ";
    cin>>f;
    Test t=f;
    t.show();
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
Enter float: 34.5
34.5
```

#12

```
#include<iostream>
using namespace std;

class Test
{
    float x;

public:
    Test(float a)
```

```
    {
        x=a;
    }
    operator float()
    {
        return x;
    }
};

int main()
{
    Test t(5.5);
    float f=t;
    cout<<f;
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
5.5
```

#13

```
#include<iostream>
#include<cmath>
using namespace std;

class Cartesian;

class Polar
{
    float r,theta;

public:
    Polar(float a,float b)
    {
        r=a;
        theta=b;
    }

    operator Cartesian();
};
class Cartesian
{
    float x,y;

public:
    Cartesian(){}

    Cartesian(float a,float b)
    {
        x=a;
        y=b;
    }

    void show()
    {
        cout<<"x = "<<x<<" y = "<<y<<endl;
    }
};
```

```
Polar::operator Cartesian()
{
    float x=r*cos(theta);
    float y=r*sin(theta);
    return Cartesian(x,y);
}
int main()
{
    Polar p(10,5);
    Cartesian c=p;
    c.show();
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
x = 2.83662 y = -9.58924
```

```
        cout<<b[i].title<<" "<<b[i].author<<"
"<<b[i].isbn<<endl;
}

int main()
{
    Library l;
    string title,author;
    long isbn;
    cout<<"Enter title, author, isbn of 5 books"<<endl;
    for(int i=0;i<5;i++)
    {
        cin>>title>>author>>isbn;
        l.addNewBook(title,author,isbn);
    }
    cout<<"Now enter isbn of book to remove: "<<endl;
    cin>>isbn;
    l.removeBooks(isbn);
    l.displayDetails();
}
```

#2

```
#include<iostream>
#include<string>
using namespace std;

class Book
{
public:
    string title,author;
    long isbn;

    Book()
    {
        title="";
        author="";
        isbn=0;
    }

    Book(string title,string author,long isbn)
    {
        this->title=title;
        this->author=author;
        this->isbn=isbn;
    }

    Book(const Book &b)
    {
        this->title=b.title;
        this->author=b.author;
        this->isbn=b.isbn;
    }
};

class Library
{
    Book b[50];
    int count;

public:
```

```
Enter 5 books (title author isbn)
b1 a1 1
b2 a2 2
b3 a3 3
b4 a4 4
b5 a5 5
Enter ISBN to remove
4
Book List
b1 a1 1
b2 a2 2
b3 a3 3
b5 a5 5
PS C:\Users\lenovo\Desktop\DSA>
```

```
    Library()
    {
        count=0;
    }
    bool addNewBook(string title,string author,long isbn)
    {
        b[count]=Book(title,author,isbn);
        count++;
        return true;
    }

    bool removeBooks(long isbn)
    {
        for(int i=0;i<count;i++)
        {
            if(b[i].isbn==isbn)
            {
                for(int j=i;j<count-1;j++)
                    b[j]=b[j+1];
                count--;
                return true;
            }
        }
        return false;
    }
    void displayDetails()
    {
        cout<<"Book List"<<endl;
        for(int i=0;i<count;i++)
            cout<<b[i].title<<" "<<b[i].author<<"
"<<b[i].isbn<<endl;
    }
};
int main()
{
    Library l;
    string title,author;
    long isbn;
    cout<<"Enter 5 books (title author isbn)"<<endl;
    for(int i=0;i<5;i++)
    {
        cin>>title>>author>>isbn;
        l.addNewBook(title,author,isbn);
    }
    cout<<"Enter ISBN to remove"<<endl;
    cin>>isbn;
    l.removeBooks(isbn);
    l.displayDetails();
}
```

#3

```
#include<iostream>
#include<string>
using namespace std;

class Account{
    const long account_number;
    long transaction_id;
    string transaction_type;
    double balance;
```

Inputs

```
a1(1,5000);
a2(2,6000);
a3(3,7000);
a4(4,8000);
a5(5,9000);
```

```
Enter from account: 2
Enter to account: 5
Enter amount: 30
```

Output

```
public:
    Account(long acc,double bal):account_number(acc){
        balance=bal;
        transaction_id=0;
        transaction_type="none";
    }
    long depositAmount(const long to,const long from,const
double amount){
        if(account_number==to){
            balance+=amount;
            transaction_type="credit";
            transaction_id++;
            return transaction_id;
        }
        return -1;
    }
    long creditAmount(const long to,const long from,const
double amount){
        if(account_number==from){
            balance-=amount;
            transaction_type="debit";
            transaction_id++;
            return transaction_id;
        }
        return -1;
    }
    void displayDetails() const{
        cout<<"Account Number: "<<account_number<<endl;
        cout<<"Balance: "<<balance<<endl;
        cout<<"Last Transaction: "<<transaction_type<<endl;
        cout<<"Transaction ID: "<<transaction_id<<endl;
        cout<<endl;
    }
};

int main(){
    Account a1(1,5000);
    Account a2(2,6000);
    Account a3(3,7000);
    Account a4(4,8000);
    Account a5(5,9000);
    long to,from; double amount; long tid;
    cout<<"Enter from account: ";
    cin>>from;
    cout<<"Enter to account: ";
    cin>>to;
    cout<<"Enter amount: ";
    cin>>amount;
    tid=a1.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a2.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a3.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a4.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a5.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;

    tid=a1.depositAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a2.depositAmount(to,from,amount); if(tid!=-1)
```

```
Transaction ID: 1

Account Details

Account Number: 1
Balance: 5000
Last Transaction: none
Transaction ID: 0

Account Number: 2
Balance: 5970
Last Transaction: debit
Transaction ID: 1

Account Number: 3
Balance: 7000
Last Transaction: none
Transaction ID: 0

Account Number: 4
Balance: 8000
Last Transaction: none
Transaction ID: 0

Account Number: 5
Balance: 9030
Last Transaction: credit
Transaction ID: 1
```

```
cout<<"Transaction ID: "<<tid<<endl;
    tid=a3.depositAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a4.depositAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a5.depositAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;

    cout<<endl<<"Account Details"<<endl<<endl;
    a1.displayDetails();
    a2.displayDetails();
    a3.displayDetails();
    a4.displayDetails();
    a5.displayDetails();
}
```

#4

```
#include<iostream>
using namespace std;

class B;
class A{
    int x;

public:
    void input(){
        cout<<"Enter value for class A"<<endl;
        cin>>x;
    }
    friend int add(A,B);
};

class B{
    int y;
public:
    void input(){
        cout<<"Enter value for class B"<<endl;
        cin>>y;
    }
    friend int add(A,B);
};

int add(A a,B b){
    return a.x+b.y;
}

int main(){
    A a;
    B b;

    a.input();
    b.input();

    cout<<"Sum = "<<add(a,b);
}
```

```
Enter value for class A
10
Enter value for class B
20
Sum = 30
```

#5

```
#include<iostream>
using namespace std;

class Complex{
    int real,imag;

public:
    Complex(){
        real=0;
        imag=0;
    }
    Complex(int r,int i){
        real=r;
        imag=i;
    }
    Complex(const Complex &c){
        real=c.real;
        imag=c.imag;
    }
    void display(){
        cout<<real<<"+"<<imag<<"i"<<endl;
    }
    friend void sum(Complex,Complex);
};
void sum(Complex c1,Complex c2){
    Complex c;
    c.real=c1.real+c2.real;
    c.imag=c1.imag+c2.imag;
    cout<<"Sum = ";
    c.display();
}

int main(){
    int r1,i1,r2,i2;
    cout<<"Enter first complex number (real imag)"<<endl;
    cin>>r1>>i1;
    cout<<"Enter second complex number (real imag)"<<endl;
    cin>>r2>>i2;
    Complex c1(r1,i1);
    Complex c2(r2,i2);
    sum(c1,c2);
}
```

```
Enter first complex number (rea
1 2
Enter second complex number (re
3 4
Sum = 4+6i
PS C:\Users\lenovo\Desktop\DSA>
```

LAB 8

FUNCTION TEMPLATE

#1

```
template <typename any>
void swapp(any &a, any &b){
	any t = a;
	a = b;
	b = t;

}
int main(){
	int x=10,y=20;
	swapp(x,y);
	cout<<"Swapped values = "<<x<<"
"<<y;
	return 0;
}
```

```
Swapped values = 20 10
--------------------------------
```

Given were 10 and 20 (ints)

```
Swapped values = g h
--------------------------
```

Given were ‘h’ and ‘g’ (chars)

#2

```
template <typename any>
int minn(int n, any a[]){
	any min=a[0];
	for(int i=1;i<n;i++){
		if(a[i]<min)
			min=a[i];
	}
	return min;
}
int main(){
	int n; int arr[10];
	cout<<"Enter size of array: ";
	cin>>n;
	for(int i=0;i<n;i++){
		cout<<"Enter element: ";
		cin>>arr[i];
	}
	cout<<"Minimum Value =
"<<minn(n,arr);
	return 0;
}
```

```
Enter size of array: 5
Enter element: 3
Enter element: 4
Enter element: 10
Enter element: -2
Enter element: -3
Minimum Value = -3
```

#3

```
template <typename any>
void bubbles(int n, any a[]){
	for(int i=0;i<n;i++){
		for(int j=0;j<n-i-1;j++){
			if(a[j]>a[j+1]){
				any temp = a[j];
				a[j]=a[j+1];
				a[j+1]=temp;
			}
		}
	}
	cout<<"Sorted Array = "<<endl;
	for(int z=0;z<n;z++){
		cout<<a[z];
	}
}
int main(){
	int n; int arr[10];
	cout<<"Enter size of array: ";
	cin>>n;
	for(int i=0;i<n;i++){
		cout<<"Enter element: ";
		cin>>arr[i];
	}
	bubbles(n,arr);
	return 0;
}
```

```
Enter size of array: 5
Enter element: 2
Enter element: 1
Enter element: 3
Enter element: 4
Enter element: 5
Sorted Array =
12345
```

#4

```
template <typename any>
int linears(int n, any a[], any val){
	for(int i=0;i<n;i++){
		if(a[i]==val)
			return i;
	}
	return -1;
}
int main(){
	int n; char arr[10]; char val;
	cout<<"Enter size of array: ";
	cin>>n;
	for(int i=0;i<n;i++){
		cout<<"Enter element: ";
		cin>>arr[i];
	}
	cout<<"Element to search for?: ";
	cin>>val;
	int pos=linears(n,arr,val);
	if (pos!=-1)
		cout<<"Element found at index:
"<<pos;
	else
		cout<<"Element not found";
```

```
Enter size of array: 4
Enter element: a
Enter element: b
Enter element: c
Enter element: d
Element to search for?: g
Element not found
```

```
Enter size of array: 5
Enter element: 3
Enter element: 2
Enter element: 4
Enter element: 5
Enter element: 6
Element to search for?: 6
Element found at index: 4
```

```
        return 0;
}
```

#5

```
void process(){
        cout<<"This is the basic overloaded function"<<endl;
}

template <typename any>
void process(any val){
        cout<<"Overloaded function process with value ="<<val<<endl;
}

int main(){
        process();
        process(3.14);
        process('a');
        return 0;
}
```

```
This is the basic overloaded function
Overloaded function process with value =3.14
Overloaded function process with value =a
```

CLASS TEMPLATE

#1

```
#include <iostream>
using namespace std;

template <typename any>
class sstack{
        any arr[10];
        int top;
        public:
                sstack(){
                        top=-1;
                }
                void push(any val){
                        if(top>=9){
                                cout<<"Stack Overflow"<<endl;
                                return;
                        }
                        arr[++top]=val;
                }
                void pop() {
    if (top < 0) {
        cout << "Stack Underflow" << endl;
        return;
    }
    top--;
                }
                void display() {
```

```
c b a
b a
```

```
9.9 7.8 1.8
7.8 1.8
```

```
        if (top < 0) {
            cout << "Stack is empty" << endl;
            return;
        }
        for (int i = top; i >= 0; i--) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};
int main(){
    sstack <char> s;
    s.push('a');
    s.push('b');
    s.push('c');
    s.display();
    s.pop();
    s.display();
    return 0;
}
```

#2

```
#include <iostream>
using namespace std;

template <class T>
class Queue {
private:
    T arr[100];
    int front, rear;

public:
    Queue() {
        front = -1;
        rear = -1;
    }
    void enqueue(T value) {
        if (rear == 99) {
            cout << "Queue Overflow" << endl;
            return;
        }
        if (front == -1) front = 0;
        rear++;
        arr[rear] = value;
    }
    void dequeue() {
        if (front == -1 || front > rear) {
            cout << "Queue Underflow" << endl;
            return;
        }
        cout << "Deleted: " << arr[front] << endl;
        front++;
    }
```

```
10 20 30
Deleted: 10
20 30
```

```
    void display() {
        if (front == -1 || front > rear) {
            cout << "Queue is empty" << endl;
            return;
        }
        for (int i = front; i <= rear; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};

int main() {
    Queue<int> q1;
    q1.enqueue(10);
    q1.enqueue(20);
    q1.enqueue(30);
    q1.display();
    q1.dequeue();
    q1.display();
    return 0;
}
```

#3

```
#include <iostream>
using namespace std;

template <class T>
class Pair {
private:
    T a, b;
public:
    void setvalue(T x, T y) {
        a = x;
        b = y;
    }
    void display() {
        cout << a << " " << b << endl;
    }
};
int main() {
    Pair<int> p1;
    p1.setvalue(67, 420);
    p1.display();
    return 0;
}
```

```
67 420
```

#4

```
#include <iostream>
using namespace std;

template <class T>
class Calc {
private:
    T a, b;

public:
    void setvalue(T x, T y) {
        a = x;
        b = y;
    }
    void add() {
        cout << "Add: " << a + b << endl;
    }
    void sub() {
        cout << "Sub: " << a - b << endl;
    }
    void mul() {
        cout << "Mul: " << a * b << endl;
    }
    void div() {
        cout << "Div: " << a / b << endl;
    }
};
int main() {
    Calc<int> c1;
    c1.setvalue(100,20);
    c1.add();
    c1.sub();
    c1.mul();
    c1.div();
    return 0;
}
```

With ints 100,20

```
Add: 120
Sub: 80
Mul: 2000
Div: 5
```

With doubles 23.2,1

```
Add: 24.2
Sub: 22.2
Mul: 23.2
Div: 23.2
```

#5

```
#include <iostream>
using namespace std;

template <class T>
class Array {
private:
    T arr[100];
    int n;

public:
    void input() {
        cout << "Enter size: ";
        cin >> n;

        for (int i = 0; i < n; i++) {
```

```
Enter size: 4
a b c d
a b c d
```

```
            cin >> arr[i];
        }
    }
    void display() {
        for (int i = 0; i < n; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};

int main() {
    Array<char> a1;
    a1.input();
    a1.display();
    return 0;
}
```

LAB 9

#1

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("NUM.txt");
    for (int i = 1; i <= 200; i++) {
        fout << i << " ";
    }
    fout.close();
    return 0;
}
```

NUM.TXT

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27
28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51
52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75
76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117
118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135
136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153
154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171
172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189
190 191 192 193 194 195 196 197 198 199 200
```

#2

```
#include <iostream>
#include <fstream>
using namespace std;

void countAlpha() {
    ifstream fin("NOTES.txt");
    char ch;
    int count = 0;
    while (fin.get(ch)) {
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
            count++;
        }
    }
    cout << "Alphabets: " << count << endl;
    fin.close();
}

int main() {
    countAlpha();
    return 0;
}
```

Notes.txt :-

```
1  jjlj
```

```
Alphabets: 4
```

#3

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream fin("NOTES.txt");
    ofstream fout("DEST.TXT");
    char ch;
    while (fin.get(ch)) {
        fout.put(ch);
```

Source

```
1  hi hii hello
```

Destination

```
1  hi hii hello
```

```
    }
    fin.close();
    fout.close();
    return 0;
}
```

#4

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    char str[100];
    int len = 0;
    cout << "Enter string: ";
    cin.getline(str, 100);

    // length
    while (str[len] != '\0') len++;
    cout << "Length: " << len << endl;

    // write to file
    ofstream fout("data.txt");
    fout << str;
    fout.close();

    // read from file
    ifstream fin("data.txt");
    char ch;
    cout << "From file: ";
    while (fin.get(ch)) {
        cout << ch;
    }

    fin.close();
    return 0;
}
```

```
Enter string: gauri gauri
Length: 11
From file: gauri gauri
```

Data.txt :-

```
gauri gauri
```

#5

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    fstream file("AZ.txt", ios::out | ios::in);

    // create A-Z file
    for (char ch = 'A'; ch <= 'Z'; ch++) {
        file.put(ch);
    }

    // (a)
    file.seekg(9);
```

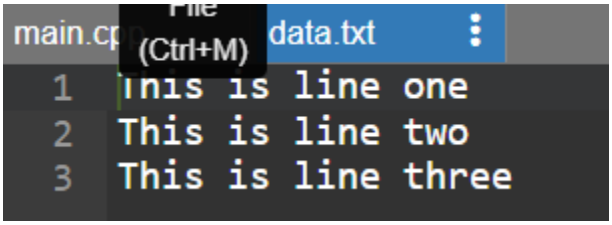
AZ.txt

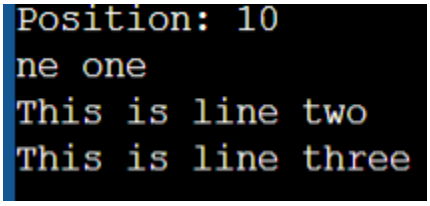
```
ABCDXFGHIJKLMNOPQRSTUVWXYZ
```

```
10th char: J
Size: 26
Last char: Z
```

```
    char ch;
    file.get(ch);
    cout << "10th char: " << ch << endl;

    // (b)
    file.seekp(4);
    file.put('X');

    // (c)
    file.seekg(0, ios::end);
    cout << "Size: " << file.tellg() << endl;

    // (d)
    file.seekg(-1, ios::end);
    file.get(ch);
    cout << "Last char: " << ch << endl;

    file.close();
    return 0;
}
```

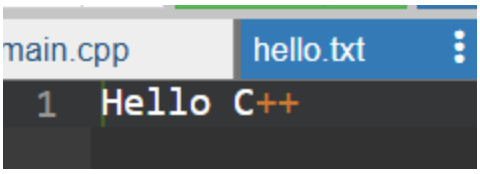
```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("data.txt");
    fout << "This is line one\nThis is line
two\nThis is line three";
    fout.close();
    ifstream fin("data.txt");

    // (i)
    fin.seekg(10);

    // (ii)
    cout << "Position: " << fin.tellg() << endl;

    // (iii)
    char ch;
    while (fin.get(ch)) {
        cout << ch;
    }
    fin.close();
    return 0;
}
```





#6

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    fstream file("hello.txt", ios::out | ios::in);
```



```
    char str[] = "Hello World";
    for (int i = 0; str[i] != '\0'; i++) {
        file.put(str[i]);
        cout << "Position: " << file.tellp() << endl;
    }
    file.seekp(6);
    file << "C++  ";  // spaces to overwrite fully
    file.close();
    return 0;
}
```

```
Position: 9
Position: 10
Position: 11
```